

Raghav Khandelwal

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Education

Washington State University

Master of Science in Computer Science

Dec 2025

Pullman, WA

- **Relevant Coursework:** Advanced Algorithms, Distributed Systems, Machine Learning, Database Systems

Experience

Community Dreams Foundation

Mar 2026 – Present

Software Developer

Remote, USA

- Developed a TypeScript/Express admin REST API for program staff, replacing 10+ hours/week of manual donor record exports with self-serve access, shipped to production across volunteer, staff, and admin roles
- Integrated Stripe and Mailchimp APIs into a donor management backend, automating recurring payment processing and email campaign triggers
- Implemented RBAC using JWT and middleware guards, enforcing permission boundaries across volunteer, staff, and admin user tiers
- Built an async background job system (Redis + Bull) for report generation and email dispatch, offloading long-running tasks and cutting UI wait time by 70%

Headstarter AI

Jun 2024 – Jul 2024

Software Engineering Fellow

Remote, USA

- Deployed a full-stack LLM-powered chatbot (Next.js, OpenAI streaming, WebSockets) achieving sub-300ms latency across 500+ concurrent sessions with JWT-secured backend
- Engineered TypeScript/Redux state layer with optimistic UI updates, reducing perceived load time by 40%

DXC Technology

Jun 2022 – Nov 2022

Associate Professional Software Engineer

India

- Designed and deployed a Python + SQL ETL pipeline extracting, transforming, and loading 6+ GB/month from DB2 into PostgreSQL, resolving integration issues between legacy and modern database environments
- Built RESTful APIs for insurance policy operations, sustaining 99.5% uptime at 500+ RPM with documented endpoint contracts for downstream consumers
- Containerized 4+ enterprise applications with Docker and configured AWS (EC2, RDS, S3), reducing environment setup time by 60% and enabling consistent deployments across dev, staging, and production
- Optimized SQL Server and DB2 query performance via index tuning and join refactoring, reducing response latency from 4.2s to 800ms, surfaced as reusable patterns for the team

Projects

Snowpack Prediction – AgAid Hackathon | Python, PyTorch, Flask, React

[Link](#)

- Designed a Transformer-based neural network for Snow Water Equivalent prediction, 87.8% accuracy across 9M+ spatio-temporal records from 8 meteorological datasets
- Engineered a scalable ingestion pipeline with chunked loading, KNN imputation, and spatial joins to ensure model reliability under variable input quality
- Deployed a Flask + React prediction platform with real-time CSV ingestion and interactive SWE visualizations

Snippetbox | Go, MySQL, net/http, Go Templates

[GitHub](#)

- Engineered a production-grade Go web server with layered middleware, panic recovery, and automated request logging, zero external runtime dependencies
- Designed a MySQL-backed repository abstraction layer, fully decoupling business logic from HTTP handlers for independent unit-testability

AI Agent | TypeScript, Node.js, Python, OpenAI API, RAG, PyTest

[GitHub](#)

- Engineered a multi-tool AI agent with RAG pipeline using OpenAI API, vector embeddings, and external API integrations
- Improved output relevance by 30% via semantic retrieval and reusable prompt configurations across varied content workflows

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Go, C++

AI & LLM: OpenAI API, RAG Pipelines, Prompt Engineering, Vector Search, Embeddings

Frameworks & Libraries: Next.js, React, Node.js, Express.js, Flask, PyTorch, Redux, Tailwind CSS

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Kubernetes, Git, GitHub Actions, CI/CD, Vercel

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, DB2